

Oral health-related quality of life and satisfaction after root canal treatment according to operator expertise: A longitudinal prospective study

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INTRODUCTION

Pulpal exposure, related pathologies, and intentional therapy account for the overwhelming majority of root canal treatments.^[1,2] With the increasing demand for

endodontic therapy, the short- and long-term effects of root canal treatment on patient quality of life (QOL) and satisfaction should be taken into consideration.^[3,4] Because tooth loss negatively affects the QOL, every measure

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should be expended to save the tooth before resorting to extraction.^[5,6]

The success rate of conventional root canal treatment is 70%–95%.^[7] However, patient satisfaction varies according to operator knowledge,^[8] skill and technique, number of appointments, duration of appointments, and whether the main symptom (typically pain) is alleviated or recurs during (flare-up) or after (complication) treatment. Because of the way in which it is perceived and processed by the brain, pain can have a lasting psychological impact and induce fear of dental treatments.

The success of endodontic treatment is evaluated based on the clinical outcome (using radiographic criteria) and the short-term survival of the tooth.^[9,10] However, these parameters alone are not sufficient to fully judge the treatment outcome.^[11] Patient-reported outcomes and satisfaction are becoming more important in modern dental practice as the patient-centered model is adopted.^[12] Moreover, patient satisfaction and QOL likely differ depending on operator expertise. The oral health-related QOL and satisfaction may be assessed using the Oral Health Impact Profile survey (OHIP-17), but studies with this instrument have focused on, for instance, malocclusion and caries.^[13–15]

We investigated the factors affecting the QOL and satisfaction of patients before and after root canal treatment performed by students, postgraduates, and specialists.

MATERIALS AND METHODS

All patients referred for root canal treatment to the outpatient wing of the dental clinics in the College of Dentistry from August to February 2019 were eligible for this prospective longitudinal study. Ethical clearance was obtained from the Local Institutional Committee for Bioethics and Research (approval no. 9-16-8/39). Sample size was calculated using the method of Liu *et al.*;^[12] 250 patients were included in the study.

All patients 18–60 years of age who were medically and mentally fit to provide informed consent were included in

the study. Patients with uncontrolled diabetes, hypertension, blood disorders, pregnancy, and undergoing orthodontic treatment and those with failed root canal-treated teeth indicated for retreatment were excluded from the study. Informed consent was obtained from all participants. Any participant who chose to withdraw from the study was allowed to do so without any consequences.

The OHIP-17 questionnaire, semantic differential scale, and radiographic assessment sheet were used.^[13] The validity of the Arabic version of the OHIP-17 was demonstrated by calculating the kappa statistic (0.95). Using the OHIP-17, an independent examiner interviewed each patient before and 2 weeks after treatment, and patient satisfaction was assessed using the differential scale 2 weeks after treatment.

Table 1: Sociodemographic distribution of patients

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Table 2: Distribution of oral health impact profile survey-17 responses among participants

Item	Frequency	Percentage	Good	Fair	Poor
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CONCLUSION

Within the limitations of this study, it can be concluded that root canal procedures improved the QOL. The findings will help dentists to understand patient perception and thus improve the holistic value of endodontic care.

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Conflicts of interest

There are no conflicts of interest.

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